NETWORK ANALYSIS

AND

SYNTHESIS

CHAPTER 1

INTRODUCTION TO NETWORKS

What are network analysis and synthesis?

- In a generally accepted definition of network analysis and synthesis, there are three key words:

  - ✓ the excitation,
  - ✓ the network, and
  - ✓ the response as depicted in Fig. 1



FIG. 1 The objects of our concern.

Cont …

- **Network analysis:** is concerned with determining the response, given the excitation and the network.
- **In network synthesis:** the problem is to design the network given the excitation and the desired response.
- In this chapter we will outline some of the problems to be encountered in this book without going into the actual details of the problems.
- We will also discuss some basic definitions.

SIGNAL ANALYSIS

- For electric networks, the excitation and response are given in terms of voltages and currents which are functions of time, t.
- In general, these functions of time are called signals.
- In describing signals, we use the two universal languages of electrical engineering
  - ✓time and
  - ✓frequency.
- Strictly speaking, a signal is a function of time.

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- However, the signal can be described equally well in terms of spectral or frequency information.

- Between time and frequency, the translation is effected by the Fourier series, the Fourier integral, and the Laplace transform.
- Consider the sinusoidal signal:

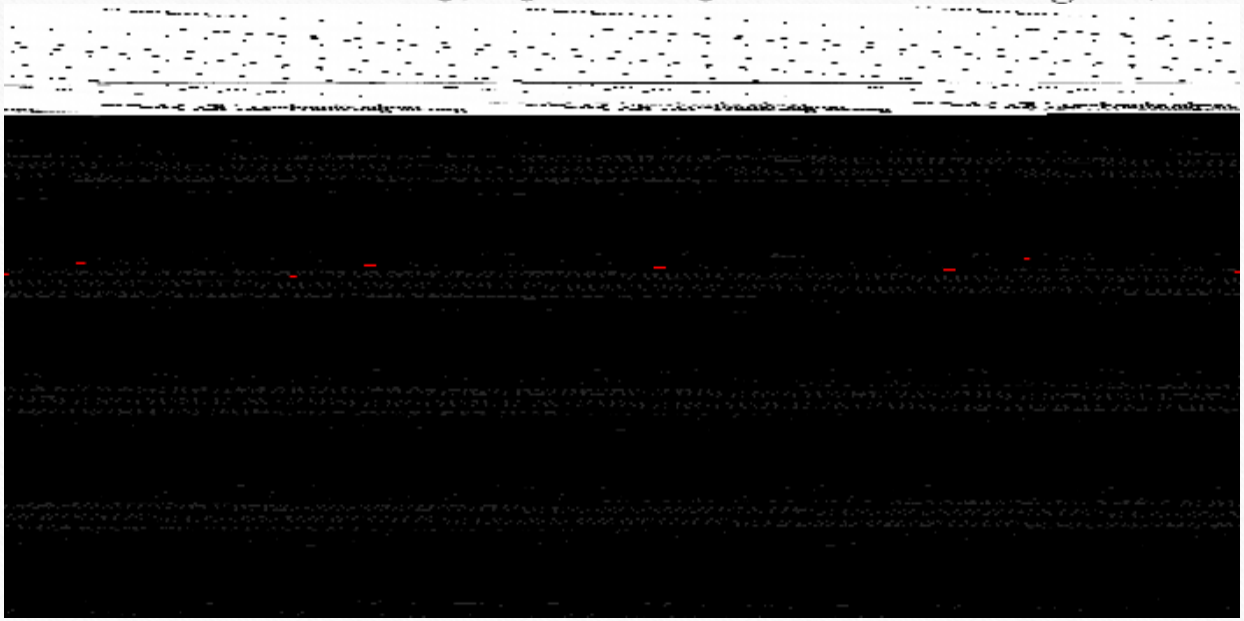
$$S(t) = A_o \sin(\omega_o t + \theta_o)$$

- where A_o is the amplitude, θ_o is the phase shift, and ω_o is the angular frequency as given by the equation:

$$\omega_o = 2\pi/T \text{ where T is the period of the sinusoid.}$$

Cont …

- The signal is plotted against time in Fig. 2.
- An equally complete description of the signal is obtained if we let the angular frequency ω be the independent variable.
- In this case, the signal is described in terms of A_o, θ_o and ω_o as shown in Fig. 3a, where amplitude is plotted against frequency, and in Fig. 1.3b, where phase shift is plotted.



Cont …

intervals $\omega_{i+1} - \omega_i$ between the lines approach zero.

- Then there is no longer any discrimination between one frequency and another, so that the discre spectra fuse into a continuous spectra, as sho the example in Figs. 5a and b.
- In the continuous case, the sum in Eq. above an integral:

$$S(t) = \int_{-\infty}^{\infty} A(\omega)\sin[\omega t + \theta(\omega)]$$

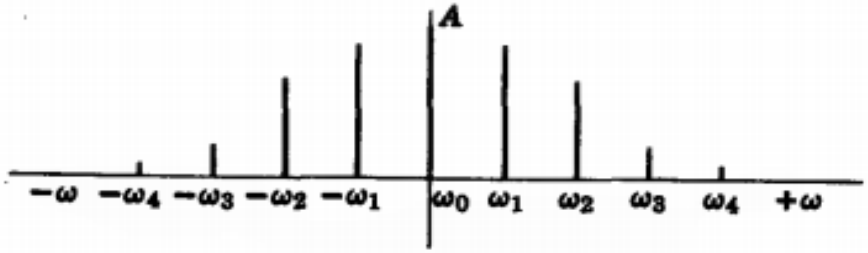


FIG. 4a. Discrete amplitude spectrum.

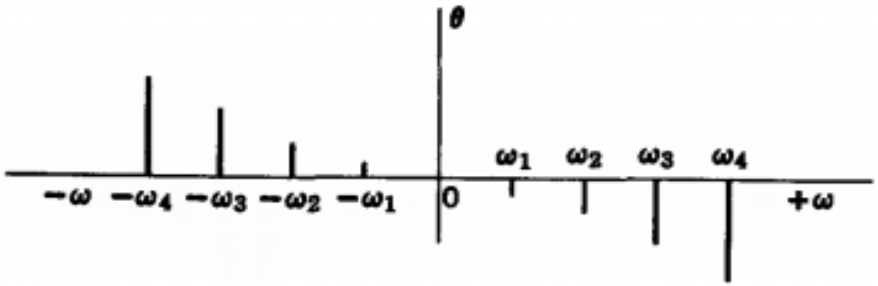


FIG. 4b. Discrete phase spectrum.

Cont …

intervals $\omega_{i+1} - \omega_i$ between the lines approach zero.

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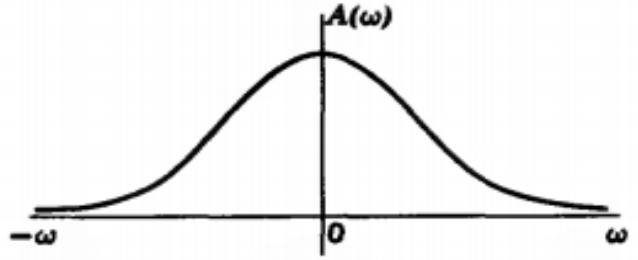


FIG. 5a. Continuous amplitude spectrum.

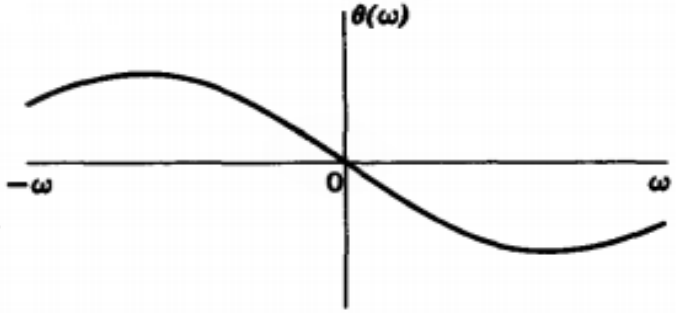


FIG. 5b. Continuous phase spectrum.

Cont …

- where A(ω) is known as the amplitude spectrum and θ(ω) as the phase spectrum.
- As we know, periodic signals such as the sine wave in Fig. 2 can be described in terms of discrete spectra through the use of Fourier series
- On the other hand, a non-periodic signal suc as the triangular pulse in Fig. 6 can only be described in terms of continuous spectra th the Fourier integral transform.

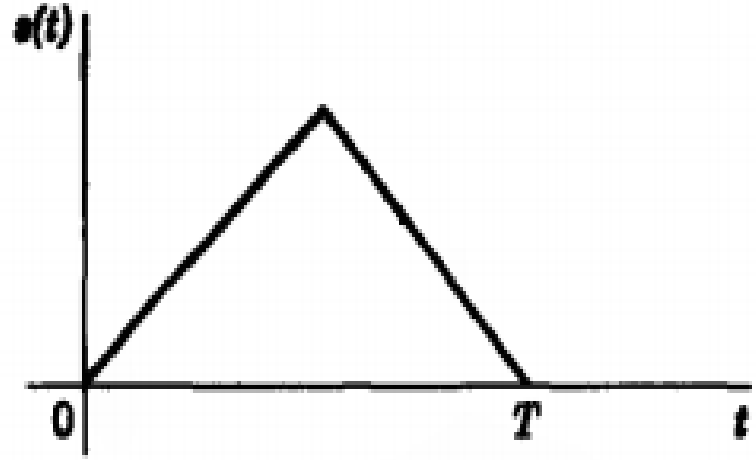


FIG. 1.6. Triangular signal.

CONCEPT OF COMPLEX FREQUENCY

- As we shall see, the complex frequency variable:

$$s = \sigma + j\omega$$

- is a generalized frequency variable whose real part $'\sigma'$ describes growth and decay of the amplitudes of signals, and whose imaginary part $'j\omega'$ is angular frequency in the usual sense.
- The idea of complex frequency is developed by
- examining the cisoidal signal:

$$S(t) = Ae^{j\omega t}$$

- when $S(t)$ is represented as a rotating phasor,

  as shown in Fig. 7.

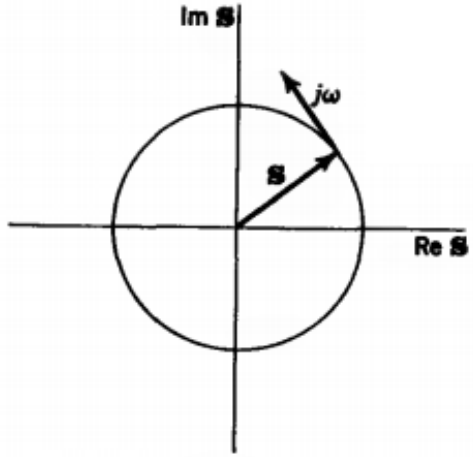


FIG. 7. Rotating phasor.

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- The resulting wave for the real and imaginary parts of S(t) are damped sinusoids as given by:

$$Re\, S(t) = Ae^{-at}\cos(\omega t)$$
$$Im\, S(t) = Ae^{-at}\sin(\omega t)$$

which are shown in Fig. 9 below.

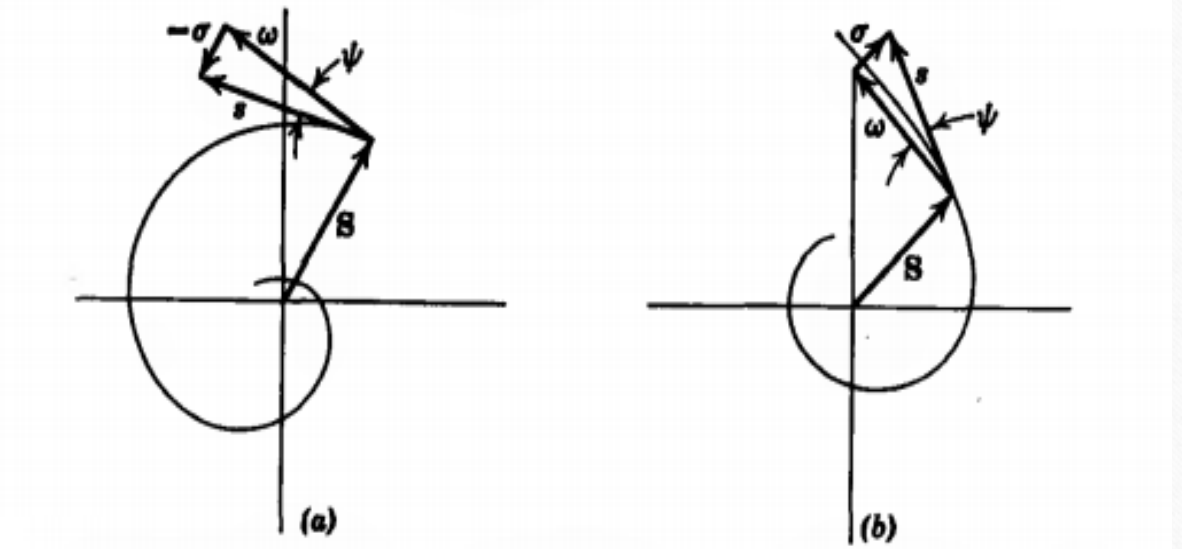
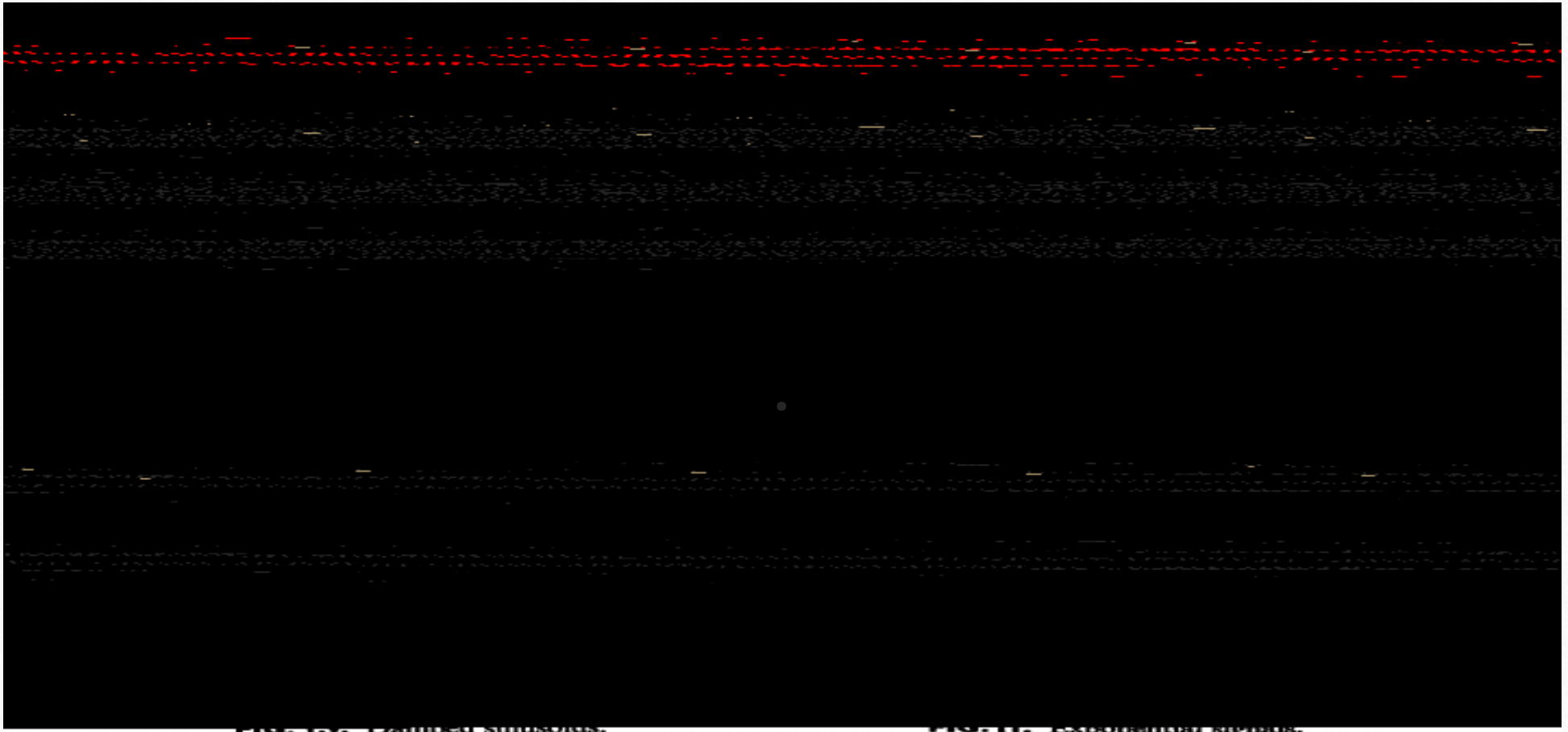


FIG. 8. (*a*) Rotating phasor with exponentially decreasing amplitude. (*b*) Rotating phasor with exponentially increasing amplitude.



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- **Proportionality principle:**
  - if for a given network, $[e(t), r(t)]$ is excitation-response pairs, then if the excitation were $e(t) = Ce(t)$, the response would be $r(t) = Cr(t)$. Where C the constant of proportionality is preserved by the linear network.

- The two conditions of linearity are summarized in Figure below.

NETWORK ANALYSIS

- As mentioned before, the characterization of the excitation and response signals in time and frequency makes up only part of the analysis problem.
- The other part consists of characterizing the network itself in terms of time and frequency, and determining how the network behaves as a signal processer.
- Let us turn our attention now to a brief study of the properties of linear networks and the general characteristics of signal processing by a linear system.

BASIC DEFINITIONS

LINEAR: A system (network) is linear if

a) the principle of superposition and

b) the principle of proportionality hold.

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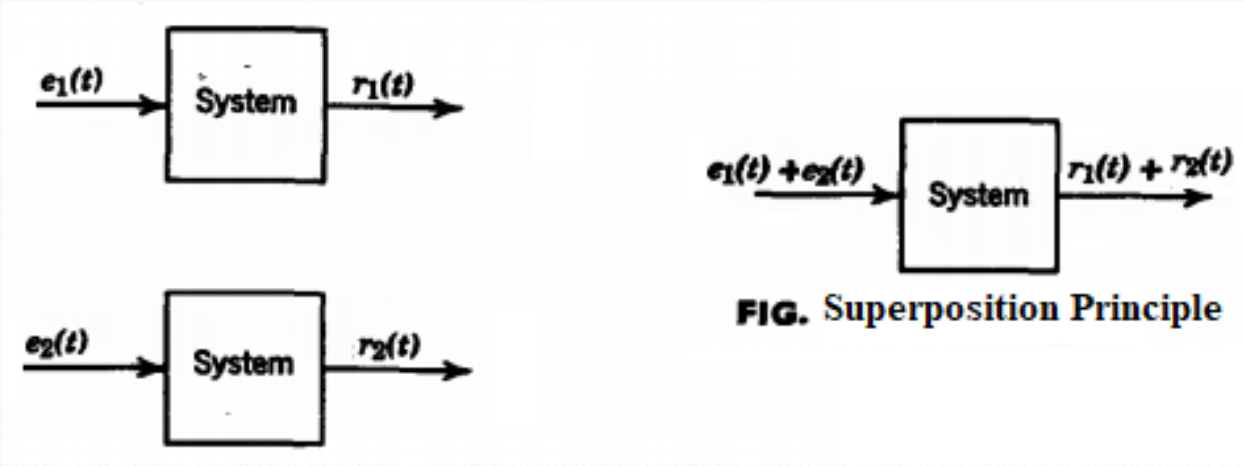
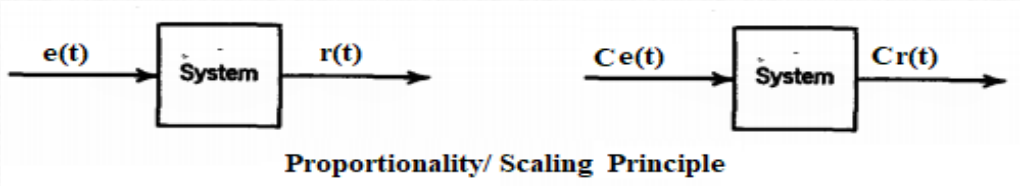


FIG. Superposition Principle

Cont …

- **Proportionality principle:**
  - if for a given network, $|e(t), r(t)|$ is excitation-response pairs, then if the excitation were $e(t) = Ce(t)$, the response would be $r(t) = Cr(t)$. Where C the constant of proportionality is preserved by the linear network.



Proportionality/ Scaling Principle

- The two conditions of linearity are summarized in Figure below.

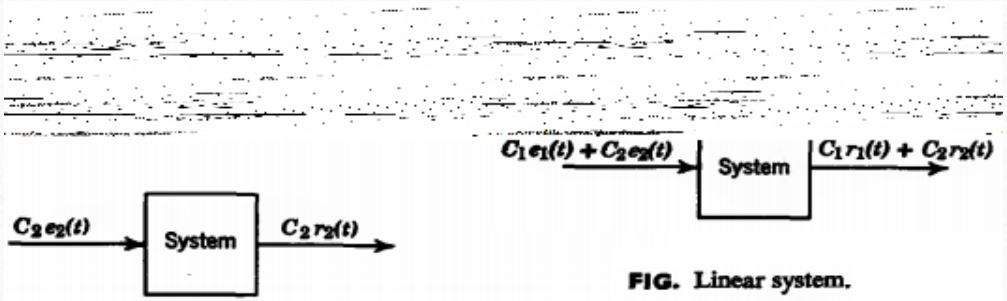


FIG. Linear system.

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- **PASSIVE**

  A linear network is passive if:

  a) the energy delivered to the network is nonnegative for any arbitrary excitation, and

  b) if no voltages or currents appear between any two terminals before an excitation is applied.

- **RECIPROCAL**

  A network is said to be reciprocal if when the points of excitation and measurement of response are interchanged, the relationship between excitation and response remains the same.

❖ Thus must be true for any choice of points of excitation and response.

- The systems given in the following all have properties which make them very useful in signal processing.

1) AMPLIFIER:

An amplifier scales up the magnitude of the input, i.e.

$$r(t) = Ke(t)$$

where K is a constant (Figure below).

➢ To understand the concept of time invariance in a linear system, let us suppose that initially the excitation is introduced at t = 0, which gives rise to a response r(t).

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- If the excitation were introduced at t = and if the shape of the response wavefo were the same as in the first case, but delayed by a time T (Figure beside), then we could say the system is time invariant.

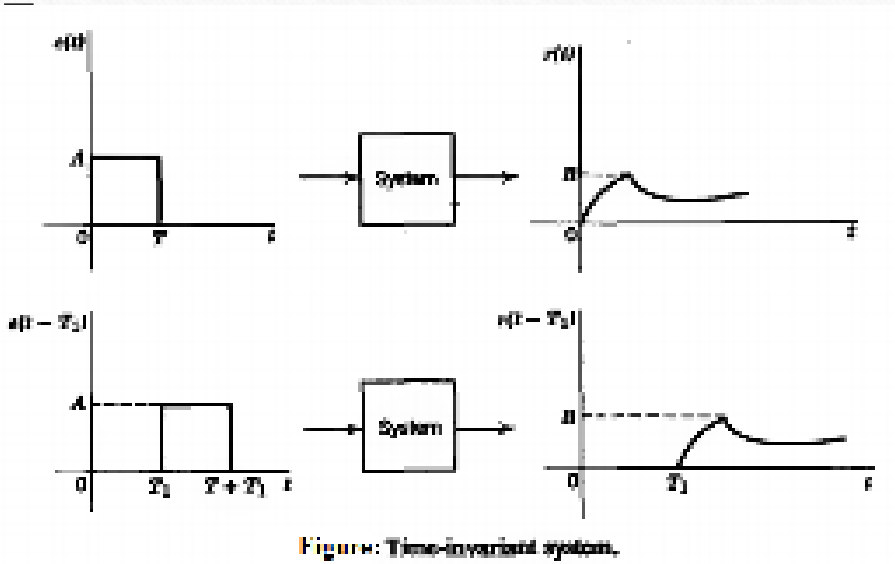


Figure: Time-invariant system.

- Another way of looking at this concept invariant systems contain only elemen
- It should be mentioned here that linear systems need not be time invariant.

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in signal processing.

1) **AMPLIFIER:**

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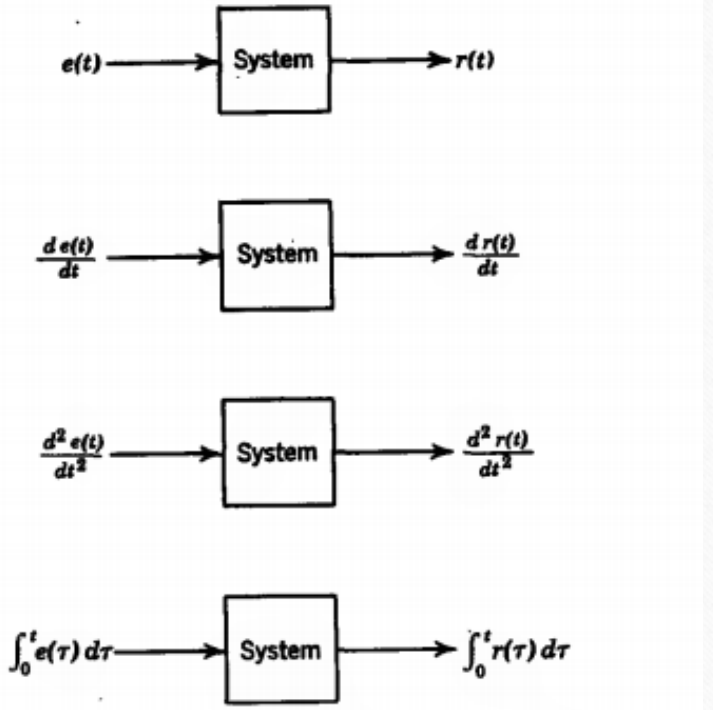
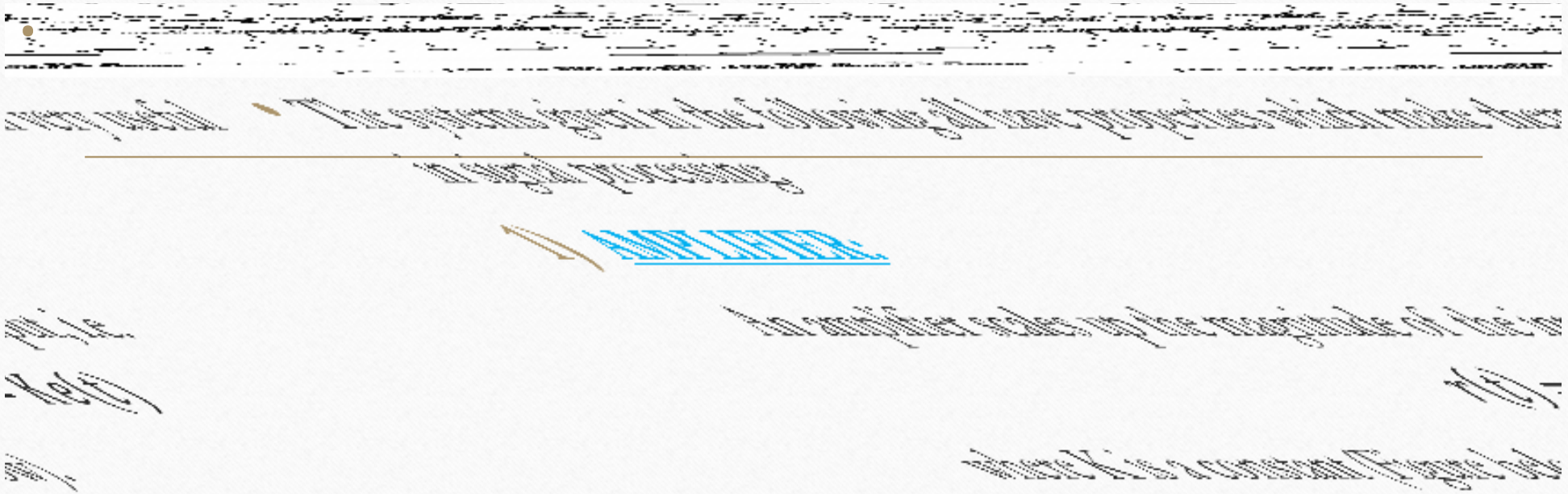


FIG. Some implications of linear time-invariant systems.

Cont …

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Ideal model

- The systems given in the following all have properties which make them very useful in signal processing.

1) AMPLIFIER:

An amplifier scales up the magnitude of the input, i.e.

$$r(t) = Ke(t)$$

where K is a constant (Figure below).



FIG. Amplifier.

Cont …

2) **DIFFERENTIATOR:** The input signal is differentiated and possibly scaled up or down (Figure below)



3) **INTEGRATOR:** The output is the integral of the input, as shown in Figure below.

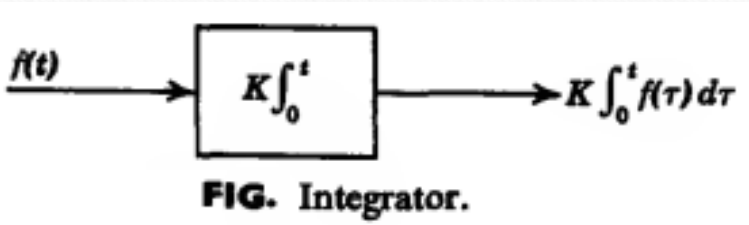


FIG. Integrator.

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4) **TIME DELAYER:** The output is delayed by an amount T, but retains the same wave shape as input (Figure below).

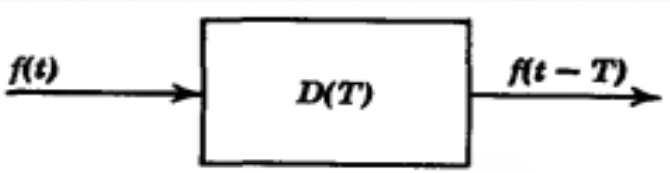
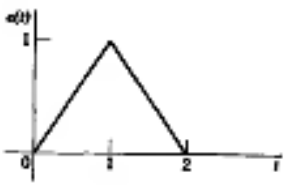
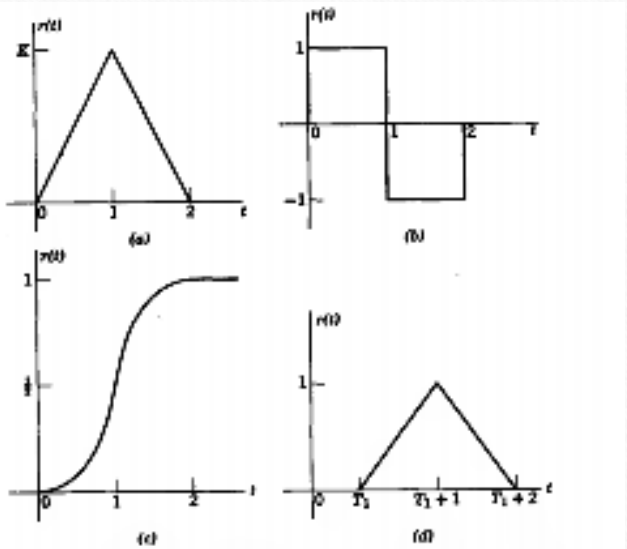


FIG. Time-delay network.

- Suppose we take the triangular puls Figure below, as the input signal.
- Then the outputs for each of the fo systems just described are shown Figures a to d.


FIG. 1.19. Excitation function.


FIG. 1.20. (a) Amplifier output. (b) Differentiator output. (c) Integrator output. (d) Delayed output.

IDEAL ELEMENTS

- In the analysis of electric networks, we use idealized linear mathematical models of physical circuit elements.
- The elements most often encountered are:
  - the resistor **R**, given in ohms,
  - the capacitor **C**, given in farads, and
  - the inductor **L**, expressed in henrys.
- The endpoints of the elements are called terminals.
- A port is defined as any pair of two terminals into which energy is supplied or withdrawn or where network variables may be measured or observed.
- In Figure below we have an example of a two-port network.

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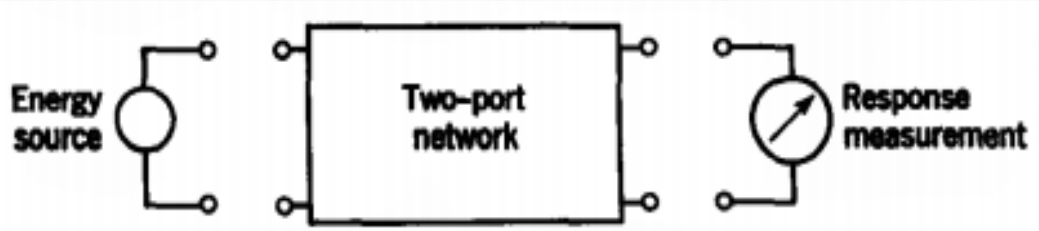


FIG. Two-port network.

- The energy sources that make up the excitation functions are ideal current or voltage sources, as shown in Figures a and b below.

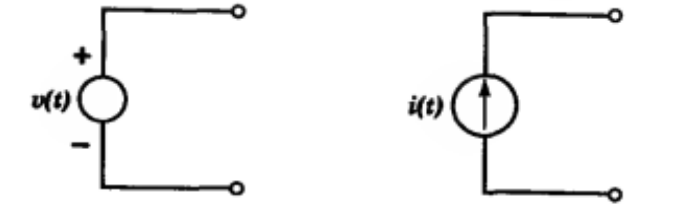


FIG. a) Voltage source. FIG. b) Current source.

- The polarities indicated for the voltage source and the direction of flow for the current source are arbitrarily assumed for reference purposes only.
- An ideal voltage source is an energy source that provides, at a given port, a voltage signal that is independent of the current at that port.

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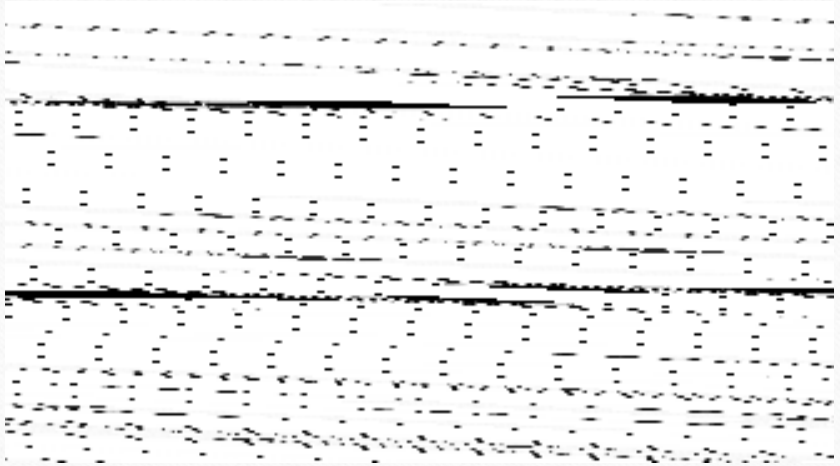
$$v(t) = \frac{1}{C}\int_0^t i(t)dx + v(0_) \quad \text{or} \quad i(t) = C\frac{d}{dt}v(t)$$

- where the constants of integration $i(0_)$ and $v(0_)$ are initial conditions to be discussed in detail later.

Cont …

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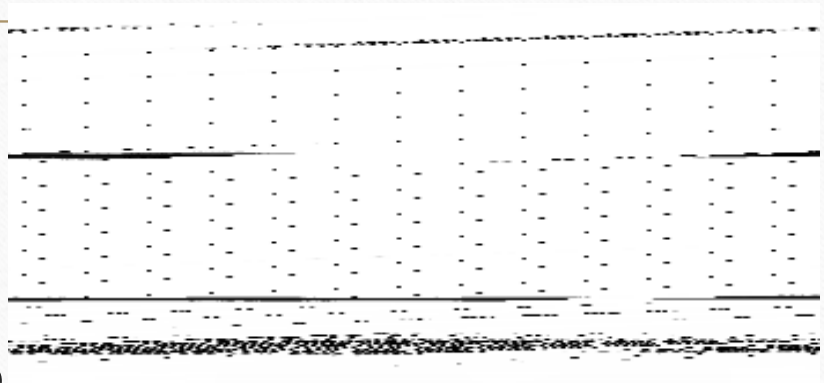
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- Expressed as a function of the complex frequency variable, s, the equations defining the R, L, and C elements, shown in Figure below, are: (ignoring initial conditions for the moment)

$$V(s) = RI(s) \quad \text{or} \quad I(s) = \frac{1}{R}V(s)$$

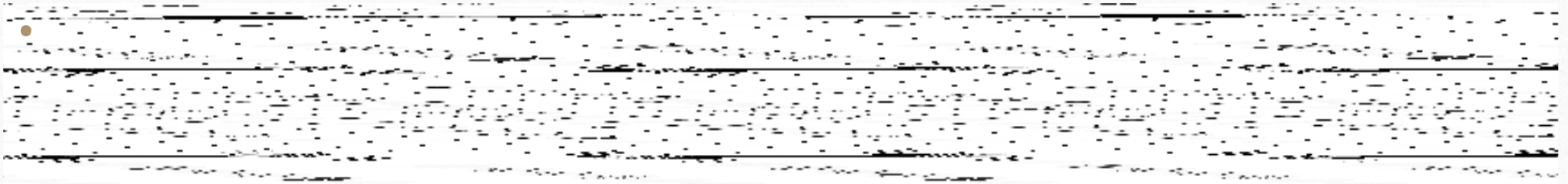
$$V(s) = sLI(s) \quad \text{or} \quad I(s) = \frac{1}{sL}V(s)$$

$$V(s) = \frac{1}{sC}I(s) \quad \text{or} \quad I(s) = sCV(s)$$



- We see that in the time domain, i.e., where the independent variable is "t", the voltage-current relationships are given in terms of **differential equations**.
- On the other hand, in the complex-frequency domain, the voltage-current relationships for the elements are expressed in **algebraic equations**.

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ample, $K = 500{,}000\Omega$ and $C = 1\mu F$ (μ = micro = 10^{-6}). or, for c

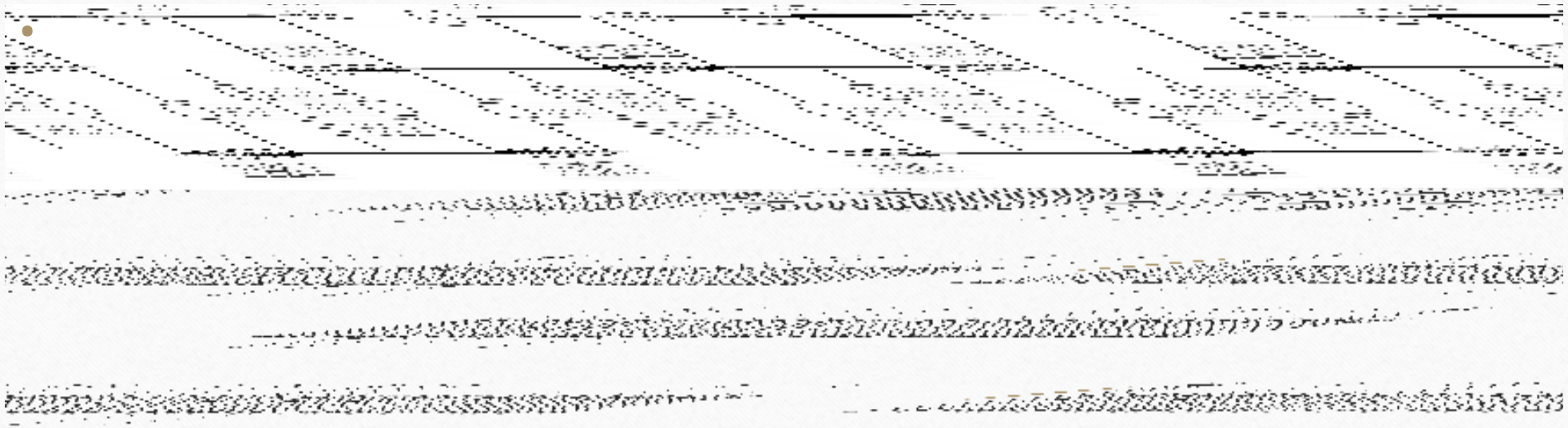
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Also we will
into high-pass,

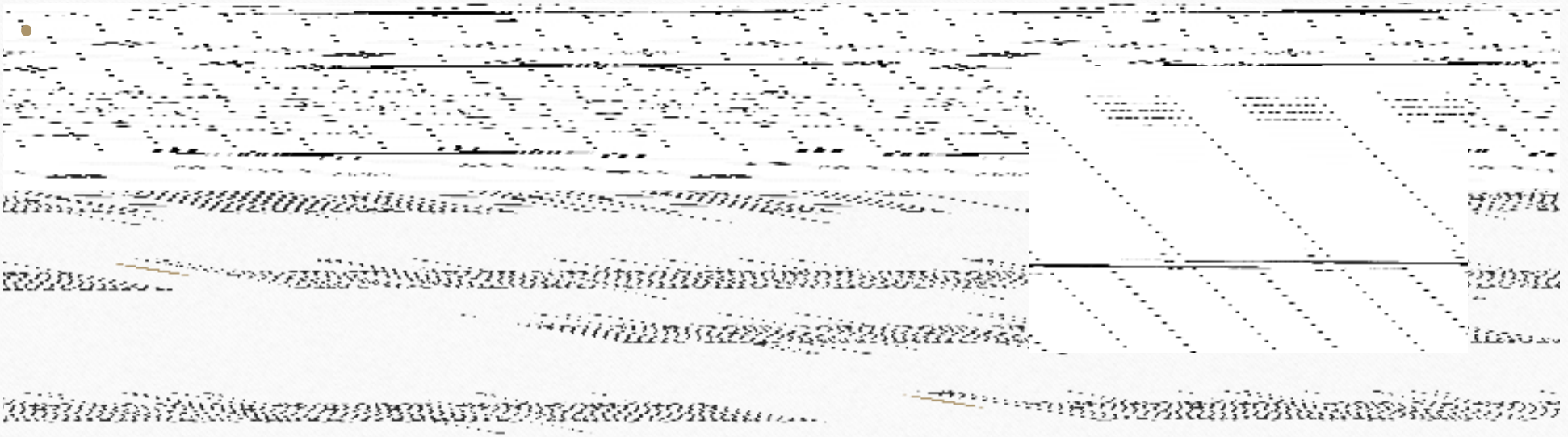
sis of this method is called frequency transformation.

The mathematical b

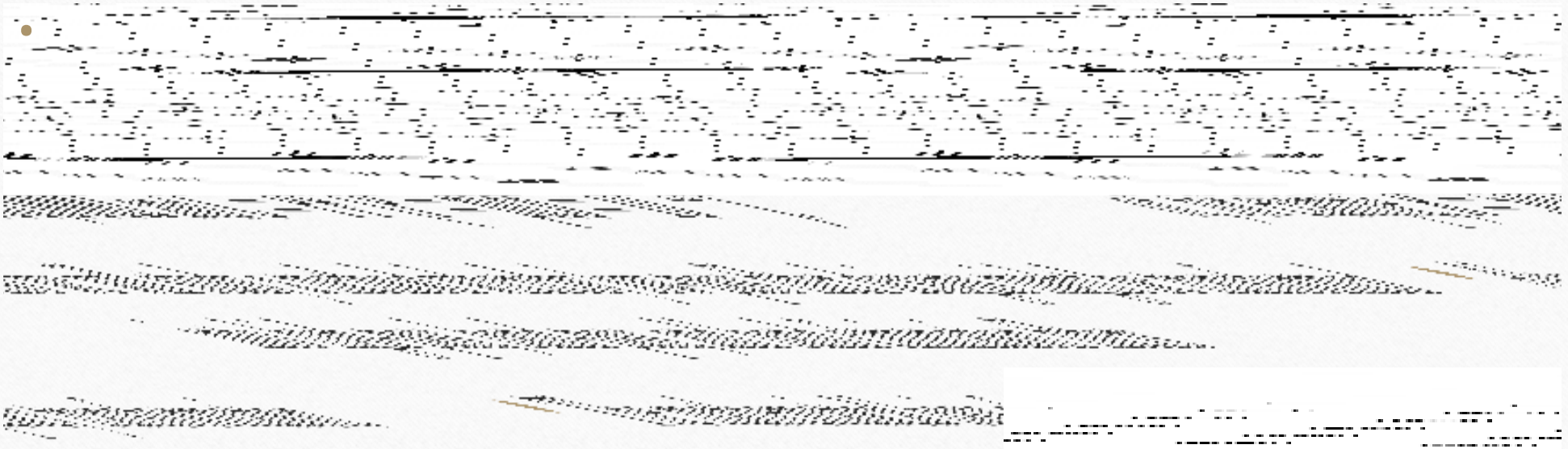
NETWORK SYNTHESIS

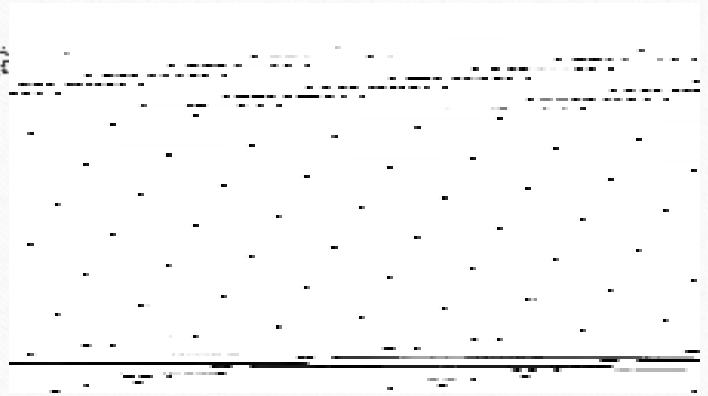


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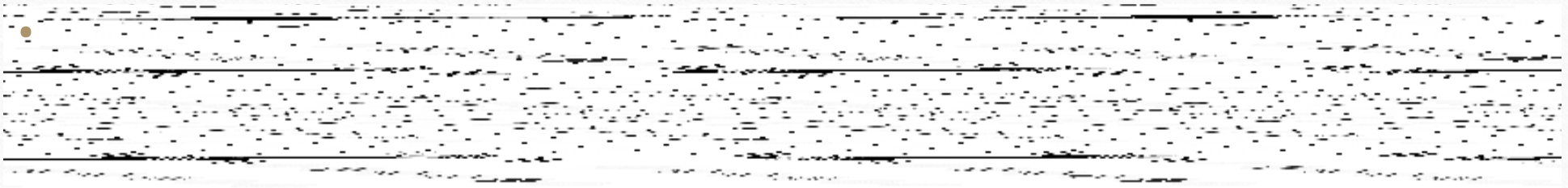


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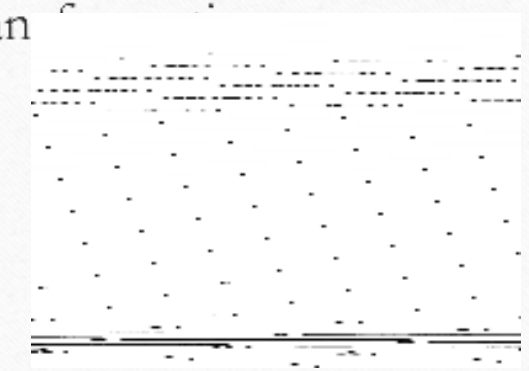


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ot, for example, $K = 500{,}000\Omega$ and $C = PF$ ($P = pico = 10^{-12}$).

- Also we will study a method whereby low-pass filter designs might be transformed into high-pass, band-pass, and band elimination filters.
- The mathematical basis of this method is called frequency tran

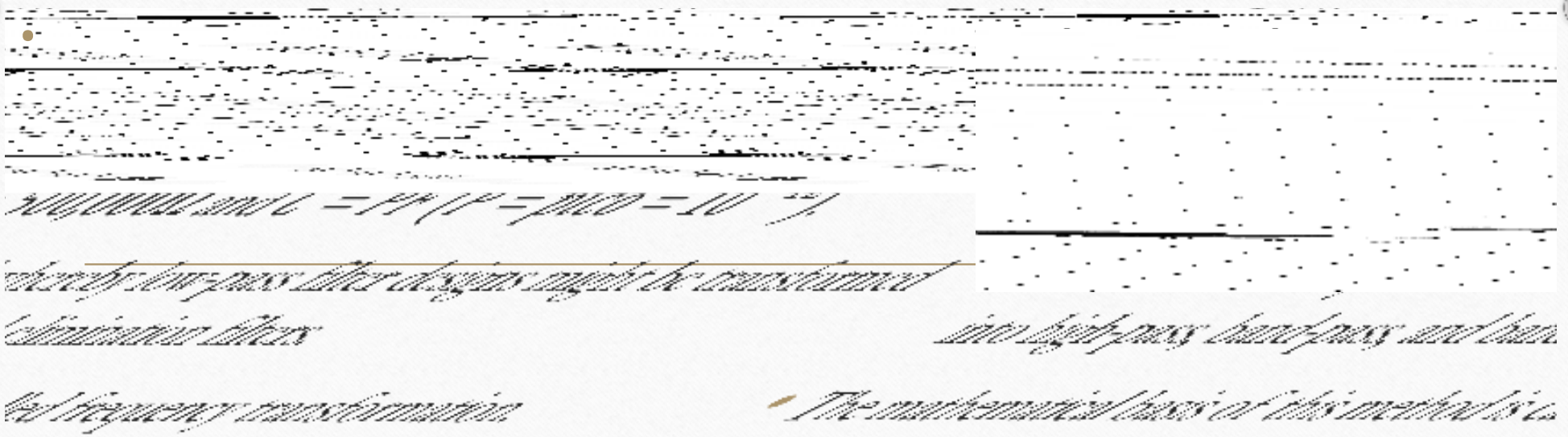


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- The problem of driving-point synthesis, as shown from the examples just given, consists of decomposing a given immittance function into basic recognizable parts (such as 3 + 4s).
- Before we proceed with the mechanics of decomposition, we must first determine whether the function is realizable, i.e., can it be synthesized in terms of positive resistances, inductances, and capacitances?
- It will be shown that realizable driving point immittances belong to a class of functions known as positive real or, simply, p.r. functions.
- From the properties of p.r. functions, we can test a given driving-point function for realizability.

Cont …

- With a knowledge of p.r. functions, we then go on to examine special driving-point functions.
- These include functions which can be realized with two kinds of elements only: the L-C, R-C, and R-L immittances.
- Next we proceed to the synthesis of transfer functions.
- A transfer function or transmittance is a system function for which the variables are measured at different ports.
- There are many different forms which a transfer function might take.
  - For example, consider the two-port network in Figure below.



some simple transfer functions.

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ample, $K = 500{,}000\Omega$ and $C = 1\mu F$ ($\mu = \text{micro} = 10^{-6}$).

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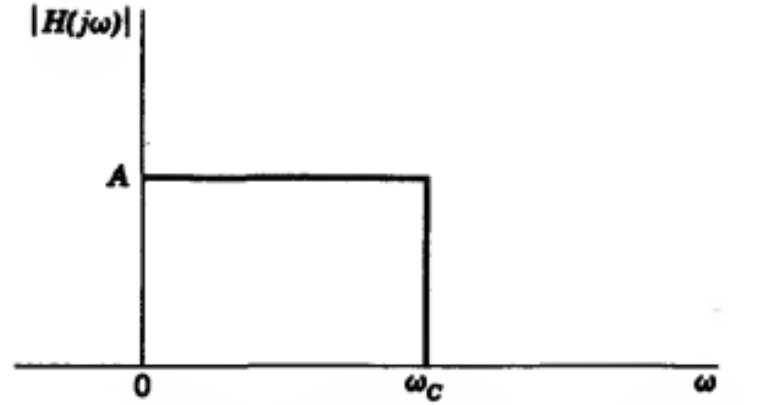
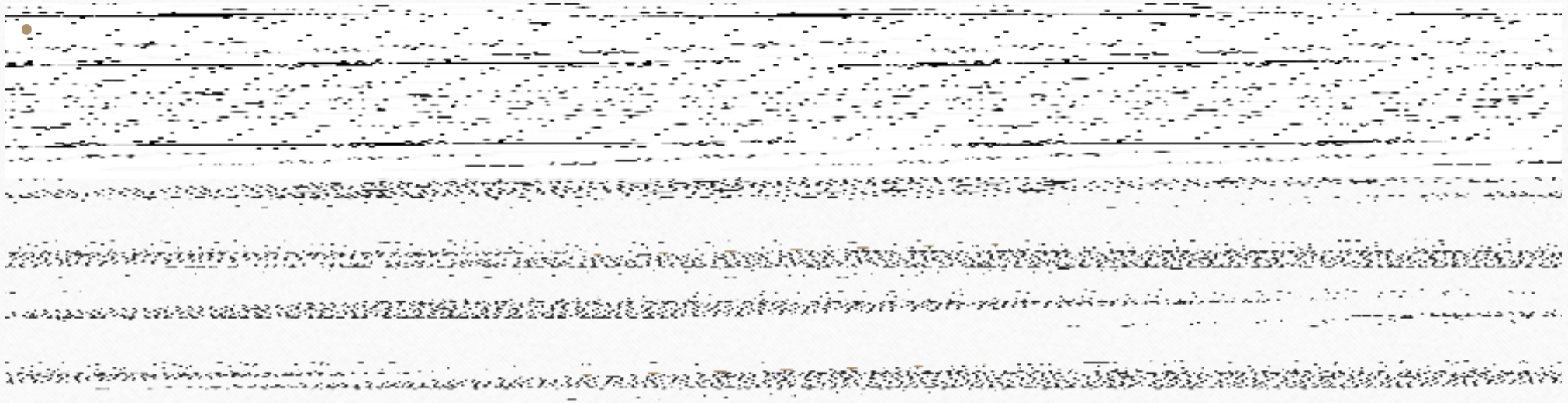
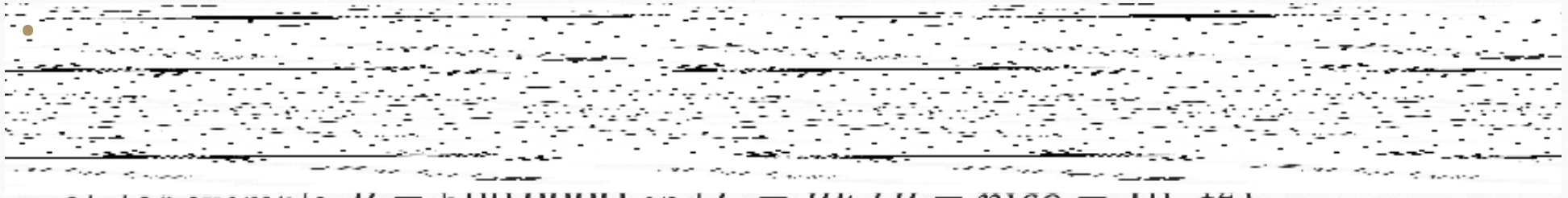


FIG. Ideal amplitude spectrum for low-pass filter.

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- Also we will study a method whereby low-pass filter designs might be transformed into high-pass, band-pass, and band elimination filters.
- The mathematical basis of this method is called frequency transformation.

Network Theorems

To analyze (Determine the output) a network, we may use one or more of the following mathematical tools (or network theorems).

1) Ohm's Law
2) Kirchhoff's Law (K-Law)
3) Superposition Theorem
4) Thevenin's and Norton's Theorem
5) Maximum Power Transfer Theorem
6) Milliman's Theorem
7) Telligen's Theorem
8) Substitution Theorem

EXERCISE: Find current in 4Ω resistor

